DIC28
True Single-Phase Clocking

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Following handout PDF pages 115–118, compare full and reduced single-phase latches, double N/P clocked inverters, and two-stage DFF chains with full/reduced first stages. An additional mid-phase input-change test addresses the requested limitations.

2 Handout requirements and test mapping

Requirement	Deck prefix and conditions	Status
Step 1: Fig. 28.1 a/c and b/d comparisons	01; independent copies	Waveforms and differences measured
Step 2: double N/P inverter hold	02; data changes while held	Hold windows measured
Q1: P/N-latch DFF, cascade two	03; 0.2/1 ns clock edges	Periodic data measured
Q2: remove first stage; limits/internal nodes	03/04; periodic and hazard data	Counterexample retained

3 Topology and expected behavior

The second stage in Fig. 28.1(a)/(b) is a full clocked inverter. Versions (c)/(d) remove the complementary-clock device that is redundant for the corresponding dynamic-node state. The full DFF combines P then N sections, giving four inversions. Removing the first stage leaves three inversions, reversing single-stage output polarity. Cascade expectations must account for this parity; first-stage waveforms cannot simply be subtracted point by point.

A single-phase architecture removes the external two-phase clock requirement but does not make internal nodes static memories. A dynamic precharge/evaluate node may support only one transition direction, and full versus reduced structures need not tolerate the same within-phase input changes.

4 Simulation setup and interpretation

The supplied encrypted MOS model is evaluated at a baseline supply of 1.0 V and 25 °C. The 1.0 V value is a teaching condition, not a verified voltage rating. These results do not reproduce the handout's original process or supply conditions. Unless stated in the deck, $W_n/W_p/L = 0.6/1.8/0.25\,\mu$m, bulks are tied to their corresponding rails, and output load is 5 fF. Transmission-gate and weak-feedback dimensions are specified explicitly in each deck.

PrimeSim HSPICE S-2021.09 win64 was used with `RUNLVL=6`, `ACCURATE`, and `WARNLIMIT=100`. Every baseline and `.alter` analysis concluded without errors or failed measurements. The simulator still reports suppressed warnings associated with encrypted blocks; this does not establish that the model is warning-free or that those warnings are harmless. The `.tran` print interval is not a fixed integration step: the solver uses adaptive timesteps. A finite sweep is not a complete process qualification.

Unless stated otherwise, the declared functional diagnostic uses $V \leq 0.2$ V for low and $V \geq 0.8$ V for high. These are report-level interpretation thresholds, not numerical handout requirements or extracted receiver noise margins. Correct samples do not exclude glitches between samples. Figures are redrawn from actual simulation data; `results/` contains per-case waveforms, measurements, and derived comparison CSV files.

Public decks omit model loading paths; model configuration is added only to local execution copies. The model itself is not distributed. The first two filename digits identify the deck numbers used below. Every deck is standalone.

5 Results

Pair	Maximum sampled voltage difference (V)
1-3	0.174150
2-4	0.043550

Paired logic agrees at the six sample times, while maximum N-section difference is 0.174150 V and P-section difference 0.043550 V. Some highs exceed 1 V and lows dip below 0 V because of floating nodes and clock feedthrough. These values must not be clipped to ideal 0/1 before claiming equivalence.

Double inverter	Hold minimum (V)	Maximum (V)	Span (V)
n	-0.052238	-0.038262	0.013977
p	1.185680	1.214528	0.028848

The N pair holds low and the P pair high, but analog levels are approximately -0.05 V and 1.2 V. Holding the intended logic and remaining at ideal supply rails are distinct properties.

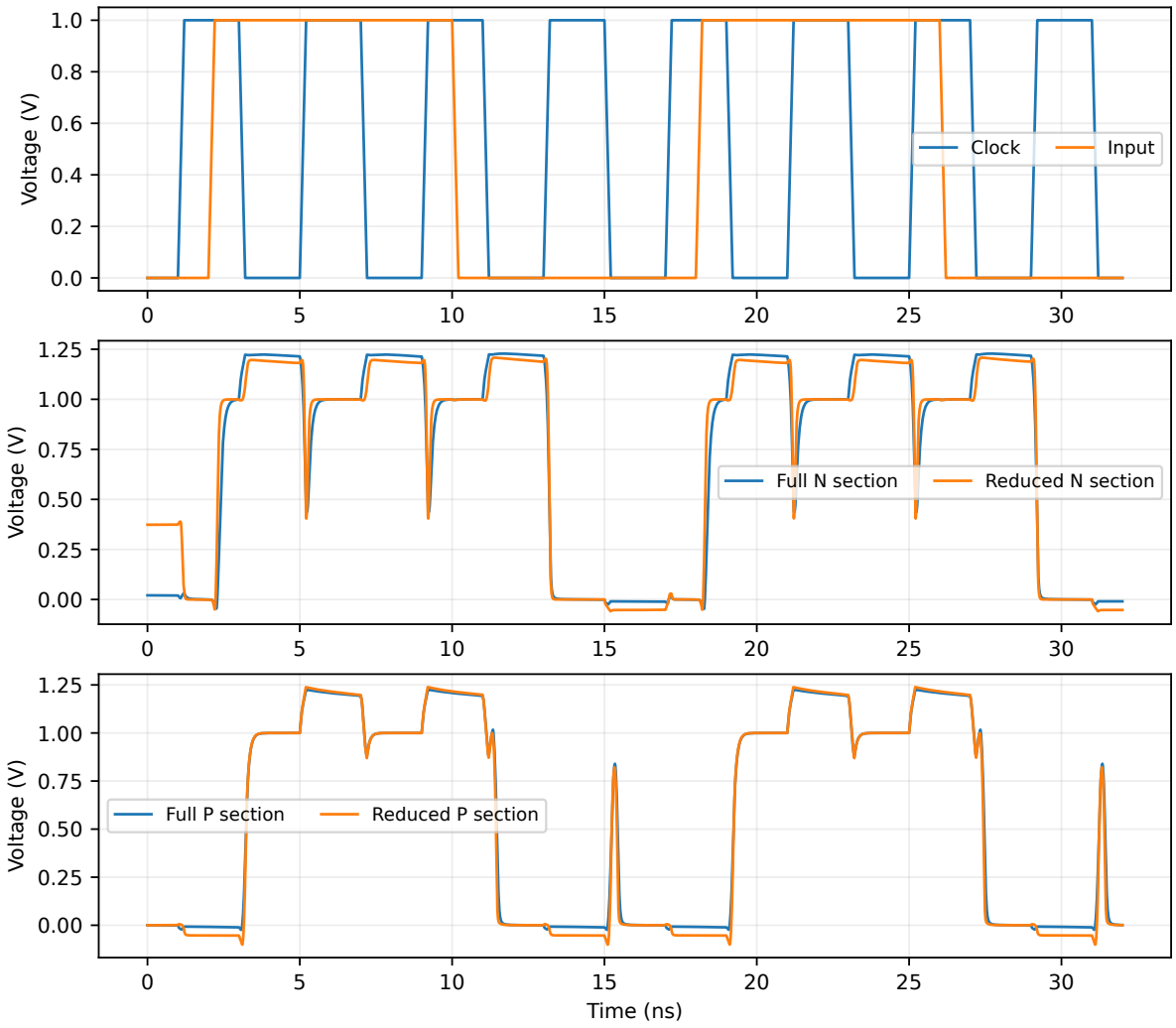


Figure 1: Paired full and simplified latches; matching logic does not mean identical analog waveforms.

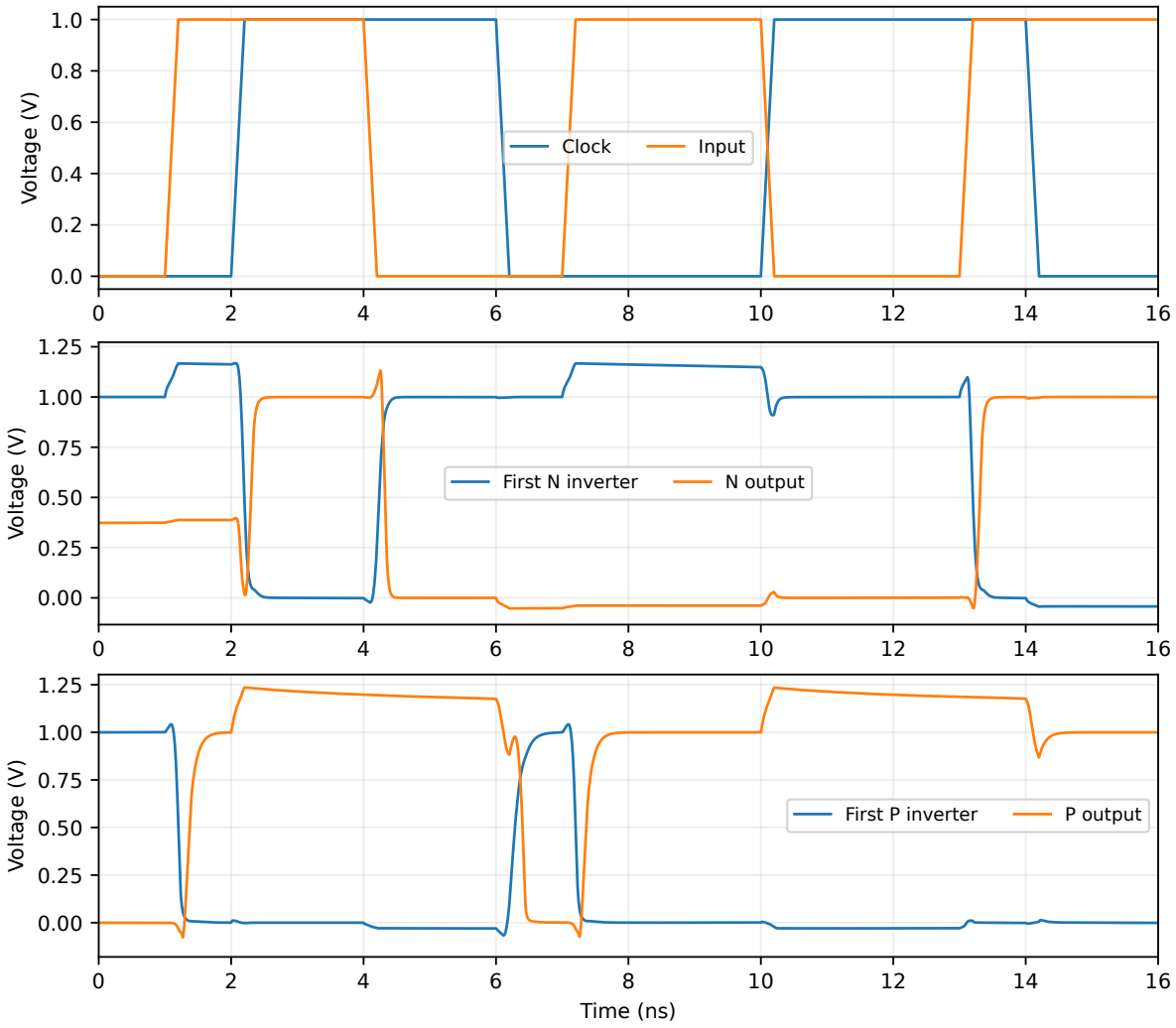


Figure 2: Input changes inside the selected hold windows: 7–9 ns for the N pair and 3–5 ns for the P pair.

Clock edge (ns)	Output	Correct samples/6
0.2	f1	6
0.2	f2	6
0.2	s1	6
0.2	s2	6
1	f1	6
1	f2	6
1	s1	6
1	s2	6

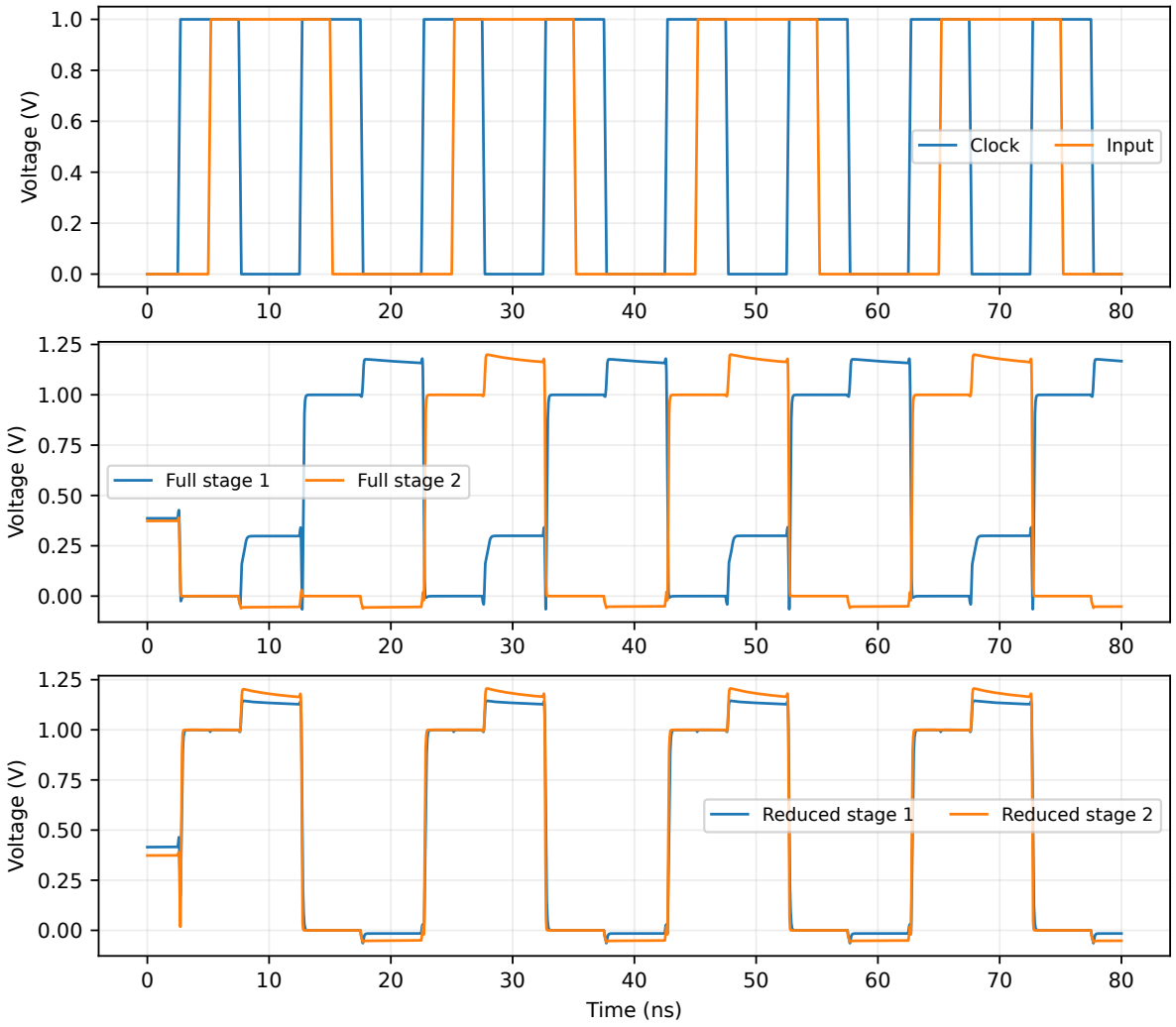


Figure 3: Two full/reduced DFF stages under periodic data; first-stage polarity differs.

With both 0.2 ns and 1 ns clock edges, all four outputs give 6/6 correct selected samples. This periodic input changes during the high phase and is settled before the next low phase begins; it does not prove acceptance of arbitrary within-phase changes.

5.1 Limitation test and internal nodes

The clock begins rising at 2.5, 12.5, and 22.5 ns. During the first high phase, input rises from 0 to 1 and returns to 0. The reduced internal node moves from 0.953449 to -0.063530 V while output remains approximately 0.999933 V: the internal node is not a complete static hold, yet completed

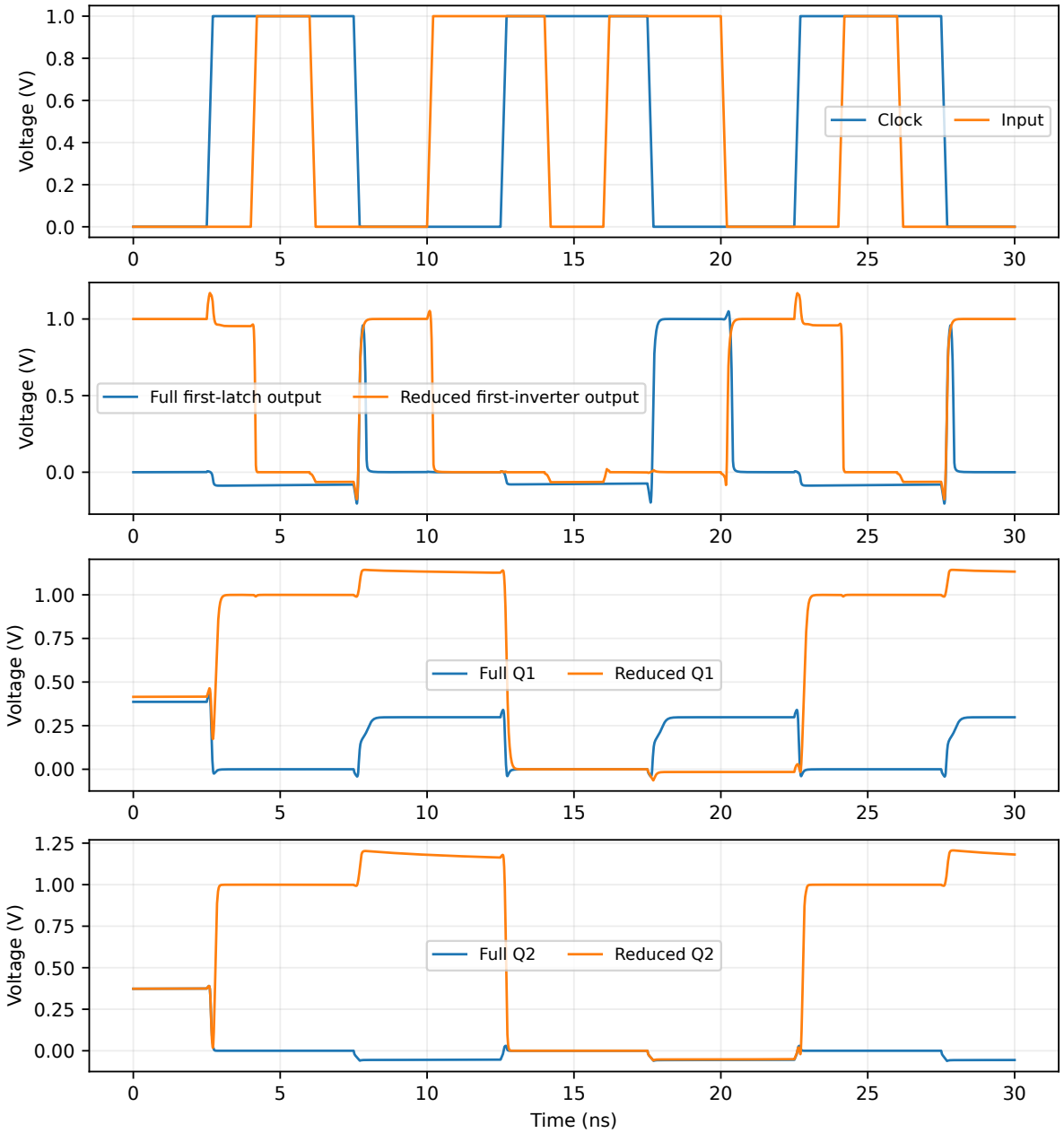


Figure 4: Additional data-timing test: the full dynamic P first stage retains previously evaluated charge, while the reduced circuit holds internal state differently.

evaluation can preserve the output. The full first-latch output moves from -0.087237 to -0.081927 V over the same interval.

In the second cycle, input changes 0 to 1 at 10 ns within the low phase. The full P-dynamic first node was already charged high and has no low-phase discharge path, so the full first-stage output remains near 0 V at 16.8 ns rather than capturing the 1 expected from a general positive-edge DFF. The reduced first stage is also near 0 V, consistent with its inverted output, and the two cascades exhibit different data sequences. This limitation counterexample preserves the handout topology; failing behavior is neither omitted nor hidden by substituting a different static DFF.

6 Discussion, differences, and unverified scope

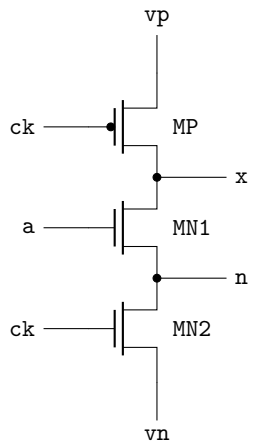
The Q1/Q2 question of correct operation requires explicit data-timing conditions: the periodic test operates, while arbitrary within-phase changes are not unconditionally equivalent to a general DFF. The full P-dynamic first stage is specifically constrained by monotonic evaluation. Although the reduced circuit preserves expected logic in the added test, partially floating internal nodes still impose setup/hold, leakage, and edge-rate limits. Six correct fixed samples are not extended into a guarantee over full hold windows, all initial states, or arbitrary frequency. The Fig. 28.5 labels 26.1(c/d) are interpreted from context as 28.1(c/d).

7 MOS transistor schematic

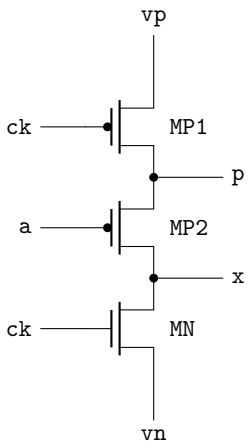
Source netlist: `01_latch_equivalence.sp; 03_two_stage_dff.sp`. Single-clock latch cells and the full/reduced flip-flop connections.

Identical node labels denote electrical connections. The drawings are editable CircuitikZ source; hierarchical boxes below refer to the MOS cells shown here.

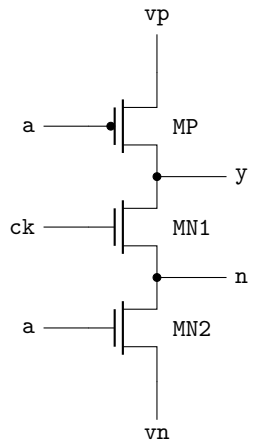
Unless explicitly labeled otherwise, every NMOS body connects to `vn`=0 and every PMOS body to `vp`=V_{DD}; body wires are omitted for readability. Ordinary inverter and dynamic/clocked cells use $W_n/L_n = 0.6/0.25\,\mu$m, $W_p/L_p = 1.8/0.25\,\mu$m. Both TG devices use $W/L = 0.6/0.25\,\mu$m.



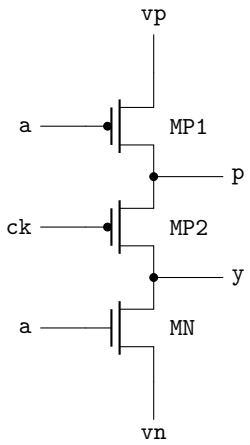
N dynamic front end (3T).



P dynamic front end (3T).

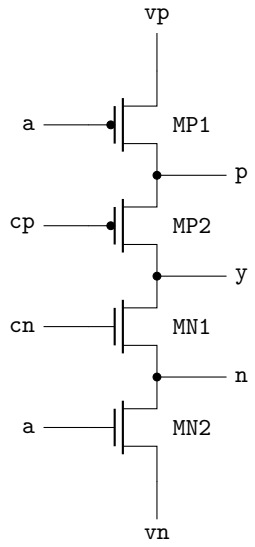


`nclock(a,y,ck,vp,vn)`: 3T



`pclock(a,y,ck,vp,vn)`: 3T

7.1 Full clocked inverter and latch hierarchy



`c2(a,y,cp,cn,vp,vn)`: 4T

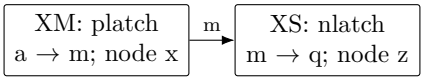
Reduced N latch (nlatch, 6T)
N dynamic front end: input a, dynamic output x, clock ck.
XO: nclock(x,y,ck).

Reduced P latch (platch, 6T)
P dynamic front end: input a, dynamic output x, clock ck.
XO: pclock(x,y,ck).
Both stages share the same clock.

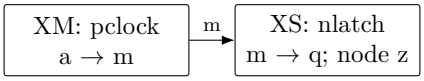
Latch test	Exact connection in the first source deck
(a), out1	N front end: MPA gate=clk, MNA1 gate=d, MNA2 gate=clk, dynamic node=a, lower node=na. XOA: c2(a,out1,cb,clk).
(b), out2	P front end: MPB1 gate=clk, MPB2 gate=d, MNB gate=clk, dynamic node=b, upper node=pb. XOB: c2(b,out2,clk,cb).
(c), out3	XNC: nlatch(d,out3,clk,c,vdd,0).
(d), out4	XPD: platch(d,out4,clk,e,vdd,0).

The generic front-end labels on the preceding page map to the instance/node names above. Versions (a)/(b) contain 7T; (c)/(d) contain 6T. Every output has 5 fF to 0, $V_{DD} = 1$ V, and $cb = V_{DD} - V(clk)$.

7.2 Two-stage DFF paths



`tfull(a,q,ck,m,x,z,vp,vn)` contains 12T: a P latch followed by an N latch. Both receive `ck`.



`tshort(a,q,ck,m,z,vp,vn)` contains 9T: the three-transistor P clocked inverter replaces the full first latch. In `03_two_stage_dff.sp`, two copies of each DFF are cascaded by connecting the first `q` to the next `a`; the clock is shared. These are the actual reduced/full circuit families used in the results, not ideal DFF symbols.